Project Flow Diagram Guide for Copilot Agent

This document provides a comprehensive instruction set and reference guide for GitHub Copilot or any AI assistant to generate visually appealing and complete Mermaid diagrams illustrating a full project flow. The goal is to produce diagrams that make the entire project easy to understand — from initiation through closure.

Project Overview

Project Name: Integrated Workflow Management System Purpose: To visualize and document the entire project lifecycle through interconnected diagrams that clearly communicate each phase, task structure, timeline, responsibility, and decision logic. Main Phases: - Initiation - Planning - Execution - Monitoring & Control - Closure

Required Diagram Types

- 1. Project Lifecycle Flowchart – Shows high-level progression (Initiation → Planning → Execution → Monitoring → Closure).
- 2. Work Breakdown Structure (WBS) – Decomposes the project into deliverables and sub-tasks.
- 3. Gantt Chart – Displays task scheduling and dependencies across a timeline.
- 4. Swimlane Diagram – Illustrates role-based responsibilities across the workflow.
- 5. Process Flow (BPMN-style) – Demonstrates task progression and approvals.
- 6. Decision Tree – Maps out project decision points and outcomes.
- 7. PERT / Network Diagram – Highlights task dependencies and the critical path.

Project Structure and Flow Logic

The diagrams must follow a logical order connecting each major project phase. Include decision points, feedback loops (e.g., Monitoring → Planning), and clear directional arrows to show flow between stages.

Roles and Responsibilities (Swimlanes)

Include the following key roles in the swimlane diagram to show how responsibilities shift throughout the workflow: - Project Manager - Developer - QA / Tester - Stakeholder

Style and Presentation Guidelines

- Use Mermaid syntax for all diagrams.
- Each diagram should be placed in its own Markdown code block.
- Prefer Left-to-Right layout for flowcharts and Top-to-Bottom for hierarchies.
- Use short but descriptive node labels for clarity.
- Add labels like 'Decision Point', 'Deliverable', and 'Feedback Loop' where relevant.

- Ensure diagrams are readable and visually balanced.

Example Mermaid Diagram

Example Code Block:

```mermaid flowchart LR A[Initiation] --> B[Planning] B --> C[Execution] C --> D[Monitoring & Control] D --> E[Closure] D --> B[Adjust Plan] ```

The Copilot agent should present each diagram under a clear section heading (e.g., '### Project Lifecycle Flow', '### Work Breakdown Structure') to make the document readable and visually cohesive. All diagrams should consistently reflect the same underlying project logic and terminology.
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